CHAPTER – 4

ANALYZING FUNCTIONAL DEPENDENCIES, REDUNDANCY AND DATA ANOMALIES

INTRODUCTION

The Amazon E-Commerce Database is designed to efficiently manage customers, sellers, products, categories, brands, carts, orders, payments, shipments, delivery partners, warehouses, inventory, reviews, wishlists, returns, and coupons. Functional dependencies identify how each non-key attribute depends on the primary key of its table, while redundancy analysis and data anomaly detection ensure that data is stored consistently and without unnecessary duplication. A well-designed database improves performance, maintains data integrity, and supports efficient management of the Amazon e-commerce platform. This chapter examines each entity of the Amazon E-Commerce Database individually, presenting its attribute structure, functional dependency, redundancy analysis, and insert, update, and delete anomalies (or the absence thereof).

1. CUSTOMER TABLE

Attribute	Key Type
Customer_ID	Primary Key
First_Name	Not Null
Last_Name	Not Null
Email	Unique Key
Phone_Number	Unique
Password	Not Null
Gender	-
Date_of_Birth	-
Address	-
City	-
State	-
Country	-
Pincode	-
Registration_Date	-

Functional Dependency

Customer_ID → First_Name, Last_Name, Email, Phone_Number, Password, Gender, Date_of_Birth, Address, City, State, Country, Pincode, Registration_Date

Redundancy Analysis

Customer information is stored only once in the Customer table. Cart, Order, Review, and Wishlist tables store only the Customer_ID, preventing duplicate customer records.

2. SELLER TABLE

Attribute	Key Type
Seller_ID	Primary Key
Seller_Name	Not Null
Business_Name	Not Null
Email	Unique Key
Phone_Number	Unique
GST_Number	Unique
Address	-
Rating	-

Functional Dependency

Seller_ID → Seller_Name, Business_Name, Email, Phone_Number, GST_Number, Address, Rating

Redundancy Analysis

Seller information is stored only once in the Seller table. The Product table stores only the Seller_ID, avoiding duplicate seller records across multiple products.

3. CATEGORY TABLE

Attribute	Key Type
Category_ID	Primary Key
Category_Name	Unique Key
Description	-

Functional Dependency

Category_ID → Category_Name, Description

Redundancy Analysis

Each category is stored only once. The Product table references the category using Category_ID instead of repeating the category name for every product.

4. BRAND TABLE

Attribute	Key Type
Brand_ID	Primary Key
Brand_Name	Unique Key
Country	-

Functional Dependency

Brand_ID → Brand_Name, Country

Redundancy Analysis

Brand details are stored only once. The Product table references the brand using Brand_ID, preventing repeated brand information across products.

5. PRODUCT TABLE

Attribute	Key Type
Product_ID	Primary Key
Product_Name	Not Null
Seller_ID	Foreign Key
Category_ID	Foreign Key
Brand_ID	Foreign Key
Price	Not Null
Discount	-
Stock	Not Null
Weight	-
Color	-
Warranty	-
Description	-
Rating	-

Functional Dependency

Product_ID → Product_Name, Seller_ID, Category_ID, Brand_ID, Price, Discount, Stock, Weight, Color, Warranty, Description, Rating

Redundancy Analysis

Product details are stored only in the Product table. Cart_Item, Order_Item, Review, Wishlist, and Inventory tables reference products using Product_ID, preventing repeated product information.

6. CART TABLE

Attribute	Key Type
Cart_ID	Primary Key
Customer_ID	Foreign Key
Total_Items	-
Grand_Total	-

Functional Dependency

Cart_ID → Customer_ID, Total_Items, Grand_Total

Redundancy Analysis

Each customer's cart summary is stored only once. Individual product-level details are kept separately in the Cart_Item table, avoiding repetition of product data in the Cart table.

7. CART_ITEM TABLE

Attribute	Key Type
CartItem_ID	Primary Key
Cart_ID	Foreign Key
Product_ID	Foreign Key
Quantity	-
Price	-

Functional Dependency

CartItem_ID → Cart_ID, Product_ID, Quantity, Price

Redundancy Analysis

The Cart_Item table stores only cart-line information. Product and cart details are referenced using foreign keys instead of being repeated for every line item.

8. ORDER TABLE

Attribute	Key Type
Order_ID	Primary Key
Customer_ID	Foreign Key
Order_Date	-
Delivery_Date	-
Order_Status	-
Total_Amount	-

Functional Dependency

Order_ID → Customer_ID, Order_Date, Delivery_Date, Order_Status, Total_Amount

Redundancy Analysis

The Order table stores only order-level information. Customer details are referenced using Customer_ID, and product-level details are kept separately in the Order_Item table.

9. ORDER_ITEM TABLE

Attribute	Key Type
OrderItem_ID	Primary Key
Order_ID	Foreign Key
Product_ID	Foreign Key
Quantity	-
Unit_Price	-

Functional Dependency

OrderItem_ID → Order_ID, Product_ID, Quantity, Unit_Price

Redundancy Analysis

The Order_Item table is maintained separately from the Order table, preventing repeated product information for orders containing multiple products.

10. PAYMENT TABLE

Attribute	Key Type
Payment_ID	Primary Key
Order_ID	Foreign Key
Payment_Method	-
Payment_Status	-
Payment_Date	-
Transaction_ID	Unique Key
Amount	-

Functional Dependency

Payment_ID → Order_ID, Payment_Method, Payment_Status, Payment_Date, Transaction_ID, Amount

Redundancy Analysis

Payment information is stored separately from the Order table to avoid duplicating order details for every payment transaction.

11. SHIPMENT TABLE

Attribute	Key Type
Shipment_ID	Primary Key
Order_ID	Foreign Key
Delivery_Partner_ID	Foreign Key
Tracking_Number	Unique Key
Shipment_Date	-
Delivery_Date	-
Shipment_Status	-

Functional Dependency

Shipment_ID → Order_ID, Delivery_Partner_ID, Tracking_Number, Shipment_Date, Delivery_Date, Shipment_Status

Redundancy Analysis

Shipment details are maintained in a separate table, preventing repeated order and delivery-partner information for every tracking update.

12. DELIVERY PARTNER TABLE

Attribute	Key Type
Delivery_Partner_ID	Primary Key
Partner_Name	Not Null
Contact_Number	-
Vehicle_Number	-

Functional Dependency

Delivery_Partner_ID → Partner_Name, Contact_Number, Vehicle_Number

Redundancy Analysis

Delivery partner information is stored only once. The Shipment table references it using Delivery_Partner_ID, avoiding duplication across multiple shipments

13. WAREHOUSE TABLE

Attribute	Key Type
Warehouse_ID	Primary Key
Warehouse_Name	-
Location	-
Capacity	-

Functional Dependency

Warehouse_ID → Warehouse_Name, Location, Capacity

Redundancy Analysis

Warehouse details are stored only once. The Inventory table references the warehouse using Warehouse_ID instead of repeating warehouse details for every stock record.

14. INVENTORY TABLE

Attribute	Key Type
Inventory_ID	Primary Key
Product_ID	Foreign Key
Warehouse_ID	Foreign Key
Available_Stock	-
Last_Updated	-

Functional Dependency

Inventory_ID → Product_ID, Warehouse_ID, Available_Stock, Last_Updated

Redundancy Analysis

The Inventory table stores only stock-level information for a Product-Warehouse pair, avoiding repetition of full product or warehouse details.

15. REVIEW TABLE

Attribute	Key Type
Review_ID	Primary Key
Customer_ID	Foreign Key
Product_ID	Foreign Key
Rating	Check (1–5)
Review_Text	-
Review_Date	-

Functional Dependency

Review_ID → Customer_ID, Product_ID, Rating, Review_Text, Review_Date

Redundancy Analysis

Review information is stored only once per review. Customer and product details are referenced using foreign keys instead of being repeated.

16. WISHLIST TABLE

Attribute	Key Type
Wishlist_ID	Primary Key
Customer_ID	Foreign Key
Product_ID	Foreign Key
Added_Date	-

Functional Dependency

Wishlist_ID → Customer_ID, Product_ID, Added_Date

Redundancy Analysis

Wishlist entries are stored only once per customer-product pair. Customer and product details are referenced using foreign keys, avoiding duplication.

17. RETURN TABLE

Attribute	Key Type
Return_ID	Primary Key
Order_ID	Foreign Key
Product_ID	Foreign Key
Return_Date	-
Return_Reason	-
Refund_Status	-

Functional Dependency

Return_ID → Order_ID, Product_ID, Return_Date, Return_Reason, Refund_Status

Redundancy Analysis

Return information is kept in a separate table, avoiding repetition of complete order and product details for every return request

18. COUPON TABLE

Attribute	Key Type
Coupon_ID	Primary Key
Coupon_Code	Unique Key
Discount_Percentage	-
Expiry_Date	-
Minimum_Order_Value	-

Functional Dependency

Coupon_ID → Coupon_Code, Discount_Percentage, Expiry_Date, Minimum_Order_Value

Redundancy Analysis

Coupon details are stored only once and can be reused across multiple orders without duplicating discount rules. In the current schema, Coupon is not yet linked to Order through a foreign key, so no cross-table redundancy exists.

CONCLUSION

The Amazon E-Commerce Database is designed using well-defined functional dependencies and normalized tables, with each entity storing only the attributes that are functionally dependent on its own primary key. Foreign keys are used throughout the schema to reference related entities instead of duplicating data, which minimizes redundancy and prevents most insert, update, and delete anomalies. A small number of update anomalies, such as recalculated cart totals or independently updated stock and status fields, are handled by confining the update to a single table. This design improves data consistency, maintains integrity, and ensures efficient database management for the Amazon E-Commerce platform.